

Project Plan

Ku CampusBite Food Delivery System

Prepared For:

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Course Title : Software Project Management

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Revision History

Version	Date	Author(s)	Summary of Changes
1.0	July 30, 2025	Cantech Team	Initial draft and baseline project plan.
1.1	Aug 16, 2025	Sana Ghulam Rasool (Project Manager)	Introduction: New section added. Project Objectives: Updated to a SMART format. Scope: Replaced "Canteen Dashboard" with "WhatsApp API Integration" to simplify order notifications for canteen staff. Also, a new "Optional Features" section was added. WBS: Revised for greater detail and accuracy.

Introduction

The KU CampusBite Food Delivery System is a software project designed to streamline the food ordering and delivery process at a large university campus. This revised project plan (Version 1.1) outlines the refined scope, updated objectives, and key deliverables for the system, which will be developed to ensure efficient and timely service for both students and faculty.

Project Objectives

1. **Project Duration:** 3 months
2. **Reduce Student Travel Costs** – Achieve up to **44% cost savings per order** by replacing physical travel to distant canteens (Rs. 90 average auto fare) with platform-based delivery (Rs. 50 or free).
3. **Save Student Time** – Ensure food delivery within **30–35 minutes** (including preparation + delivery), reducing the current **45–60 minutes of commuting and queuing**.
4. **Increase Accessibility & Reach** – Provide a **centralized ordering platform for all 15 canteens**, enabling students easy access while helping canteens **expand customer reach by 10–20%**.
5. **Enhance Efficiency** – Improve order processing speed by **20% through automation**, reducing manual errors and waiting times.
6. **Ensure Student Satisfaction** – Achieve **60–65% positive feedback** on usability, convenience, and reliability within the **first 3 months** of implementation.

Project Scope

Included:

- 1 Features for meal pre-ordering, order-scheduling from multiple KU canteens.
- 2 Secure user and admin login, plus guest access without account creation.
- 3 Student facing interface for browsing menus, searching items, managing carts, and placing orders.
- 4 Order details automatically sent to the respective canteen via Whatsapp.
- 5 Mobile-responsive website design for easy access from smartphones and laptops.
- 6 Onboarding of canteen staff with guidance on using the system and updating item availability status.

Excluded:

- 1 Payment gateway integration - initial version will use cash on delivery.
- 2 Integration with student ID cards or university academic systems.
- 3 Off-campus deliveries – service is strictly for Karachi University campus.
- 4 Multilingual interface – initial version will be in English only.
- 5 Native mobile application – the system will be web-based only.
- 6 Real-time delivery tracking on a map.

Optional Features

- 1 Menu filtering – price-based and canteen-based filtering options
- 2 Ratings and feedback – user reviews and ratings for canteens and orders

- 3 Menu availability update tool – online tool for canteen staff to update item availability
- 4 Help/FAQ section – quick-start guide for students on using the website

Work Breakdown Structure (WBS)

1. Project Initiation & Documentation

- 1.1 Define project objectives and goals
- 1.2 Identify key stakeholders and roles
- 1.3 Prepare initial project charter and scope outline
- 1.4 Documentation (maintaining all key project documents and updating them as needed throughout lifecycle)

2. Requirements Gathering & Analysis

- 2.1 Gather requirements from stakeholders
- 2.2 Identify functional requirements
- 2.3 Identify non-functional requirements (usability, availability, etc.)

3. Design

- 3.1 UI/UX Design (basic interface structure, initial layout decisions)
- 3.2 Database Design (ER diagram, schema notes)
- 3.3 Use Case Diagram (system interaction representation)

4. System Development

- 4.1 Student Interface Development
- 4.2 Login & Authentication
- 4.3 Admin Dashboard Development
- 4.4 Backend & Frontend Integration
- 4.5 WhatsApp Integration for canteen notifications

5. Testing & Quality Assurance

- 5.1 Unit Testing (testing each module individually)
- 5.2 Integration Testing (testing how modules work together — e.g., login + student interface + database)
- 5.3 System Testing (end-to-end testing of the entire application)
- 5.4 User Acceptance Testing (stakeholders test if the system meets needs)
- 5.5 Test additional features after implementation

6. Enhancements & Improvements

- 6.1 Implement additional features (if time/resources allow)
- 6.2 Improve UI based on testing and feedback
- 6.3 Re-test enhancements

7. Deployment & Launch

- 7.1 Prepare deployment environment
- 7.2 Deploy final application
- 7.3 Official project launch

Schedule and Milestones

Schedule:

Activity/Phase	Start Date	End Date
Roles and Requirements Finalization	25 July 2025	31 July 2025
UI Design and Database Schema Completion	1 Aug 2025	7 Aug 2025
Core Frontend and Backend Development	8 Aug 2025	18 Aug 2025
Student Order Flow Implementation	19 Aug 2025	26 Aug 2025
Whatsapp Integration	27 Aug 2025	2 Sept 2025
Admin Dashboard Development	3 Sept 2025	15 Sept 2025
Integrated Testing and User Feedback Collection	16 Sept 2025	24 Sept 2025
Final bug fixes and UI/UX Polishing	25 Sept 2025	2 Oct 2025
Final Deployment	3 Oct 2025	8 Oct 2025

Milestones:

Milestone	Target Date
Requirements Finalization	31 July 2025
UI Design & Core Backend Structure Implemented	18 Aug 2025
Development Completed	15 Sept 2025
Integrated Testing	24 Sept 2025
Final Deployment	8 Oct 2025

Roles and Resources

Role/Resource	Person/Team
Project Manager	Sana Ghulam Rasool
Business Analyst	Sania Khan
Design Team/Frontend Development	Sania Khan, Humaima Moin, Lamia Gull
Backend Development Team	Neha Rehan, Sana Ghulam Rasool, Lamia Gull
Quality Assurance Team	Sana Ghulam Rasool, Neha Rehan
End User Representatives	Classmates (Students)
DevOps Team	Lamia Gull, Sania Khan, Humaima Moin

Risk Identification and Mitigation

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1	Internet Connectivity Issues Some parts of the university may have weak Wi-Fi or mobile signals, making it hard for users to place orders or receive real-time updates.	High	Build offline order support with auto-sync and use SMS fallback for critical updates.
2	Server Downtime If the app server goes down, no one can place or receive orders.	High	Host on a reliable cloud service with 99.9% uptime and enable auto-recovery with admin alerts.
3	Data Privacy Concerns Students' personal information (e.g., phone numbers, location) may be misused.	High	Encrypt sensitive data, limit staff access to essentials, and provide a privacy policy with user control.
4	Vendor Overload During peak hours, vendors might receive more orders than they can manage.	Medium	Set order limits per vendor and let them mark themselves as temporarily unavailable.
5	Food Wastage or Spoilage If a student doesn't pick up their food, it may spoil or be wasted.	Low	Set a 10-minute pickup window with order reassignment and apply upfront payments or penalties for no-shows.

Communication Plan

Purpose	Who will they talk	How they will talk	When
Share the project progress	Cantech team	Weekly meeting	Every Monday
Design and Feature discussions	Developer	WhatsApp or a meeting	2 times a week or as needed
Ask for feedback	Stakeholders	Survey	During the testing phase
Report problems or bugs	Developers	WhatsApp	As soon as the issue happens
Final update and project delivery	Cantech and developers	Presentation or meeting	At the end of the project

Change Control Process

Steps	What will happen
Request Change	A team member, student, or supervisor suggests a change
Record the Change	Team writes it down in a change log
Review the Change	Team checks that it is useful, can we do it in time? Does it affect other parts?
Approve or Reject	PM and team decide to approve or reject
Make the Change	A developer or a designer makes the change in the system
Inform everyone	Team informs all members and updates documentation